MARKING SCHEME & STEP-WISE MODEL ANSWERS

CLASS 10 MATHEMATICS UNIT TEST - QUADRATIC EQUATIONS

CLASS 10 — MATHEMATICS (MAX MARKS: 29)

SECTION A (Multiple Choice & Objective Questions - 1 Mark each)

Q1. (Quadratic Equations) **[1 Marks]**

Assertion (A): The equation

$$x^2 + 3x + 4 = 0$$

has no real roots. Reason (R): For a quadratic equation

$$ax^2 + bx + c = 0$$

, if

$$b^2 - 4ac < 0$$

, the roots are imaginary.

Correct Answer: Option (A)

Model Answer:

$$D = 3^2 - 4(1)(4) = 9 - 16 = -7 < 0$$

. Since discriminant is negative, roots are imaginary. Both A and R are true and R explains A.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Evaluating discriminant D = −7 < 0 and validating assertion and reason	1.0 M

Q2. (Real Numbers) **[1 Marks]**

The exponent of

$$2$$

in the prime factorisation of

$$144$$

is:

Correct Answer: Option (A)

Model Answer:

Prime factorisation of

$$144 = 2^4 \times 3^2$$

. The exponent of

$$2$$

is

$$4$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Correct prime factorisation and identification of power of 2	1.0 M

Q3. (Quadratic Equations) **[1 Marks]**

If the discriminant of equation

$$ax^2 + bx + c = 0$$

is zero, then the roots are:

Correct Answer: Option (A)

Model Answer:

When

$$D = b^2 - 4ac = 0$$

, both roots are

$$-\frac{b}{2a}$$

, so they are real and equal.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Stating D = 0 implies real and equal roots	1.0 M

Q4. (Quadratic Equations) **[1 Marks]**

If the equation

$$x^2 - kx + 9 = 0$$

has equal roots, then the value of

$$k$$

is:

Correct Answer: Option (A)

Model Answer:

For equal roots,

$$D = (-k)^2 - 4(1)(9) = 0 \Longrightarrow k^2 - 36 = 0 \Longrightarrow k = +-6$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Setting D = k^2 - 36 = 0 to get k = +-6	1.0 M

Q5. (Quadratic Equations) **[3 Marks]**

The sum of the reciprocals of Rehman's ages (in years)

$$3 \text{ years ago}$$

and

$$5 \text{ years from now}$$

is

$$\frac{1}{3}$$

. Find his present age.

Model Answer:

Let Rehman's present age be

$$x$$

years. 3 years ago age was

$$x - 3$$

, 5 years from now age is

$$x + 5$$

.

$$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3} \implies \frac{2x+2}{x^2+2x-15} = \frac{1}{3} \implies 6x + 6 = x^2 + 2x - 15 \implies x^2 - 4x - 21 = 0$$

. Factoring:

$$(x-7)(x+3) = 0 \implies x = 7$$

(age cannot be negative). Rehman's present age is

$$7 \text{ years}$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Forming reciprocal equation 1/(x-3) + 1/(x+5) = 1/3	1.0 M
2	Simplifying to x^2 - 4x - 21 = 0	1.0 M
3	Solving to find present age x = 7 years	1.0 M

SECTION B (Very Short Answer Questions - 2 Marks each)

Q6. (Quadratic Equations) **[2 Marks]**

Find the values of

$$k$$

for which the quadratic equation

$$kx(x-2) + 6 = 0$$

has two equal real roots.

Model Answer:

Rewriting equation:

$$kx^2 - 2kx + 6 = 0$$

. Here

$$a = k, b = -2k, c = 6$$

. For equal roots,

$$D = 0 \implies (-2k)^2 - 4(k)(6) = 0 \implies 4k^2 - 24k = 0 \implies 4k(k-6) = 0$$

. Since

$$k \neq 0$$

(otherwise not quadratic),

$$k = 6$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Rewriting in standard form and setting D = b^2 - 4ac = 0	1.0 M
2	Solving 4k(k-6)=0 and rejecting k=0 to get k = 6	1.0 M

Q7. (Quadratic Equations) **[3 Marks]**

A motor boat whose speed is

$$18 \text{ km/h}$$

in still water takes

$$1 \text{ hour}$$

more to go

$$24 \text{ km}$$

upstream than to return downstream to the same spot. Find the speed of the stream.

Model Answer:

Let stream speed be

$$x \text{ km/h}$$

. Upstream speed

$$= 18 - x$$

, Downstream speed

$$= 18 + x$$

. Time difference:

$$\frac{24}{18-x} - \frac{24}{18+x} = 1 \implies 24\left(\frac{18+x-18+x}{18^2 - x^2}\right) = 1 \implies 48x = 324 - x^2 \implies x^2 + 48x - 324 = 0$$

. Factoring:

$$(x + 54)(x - 6) = 0 \implies x = 6$$

(speed cannot be negative). Speed of stream is

$$6\ \mathrm{km/h}$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Setting up time equation 24/(18-x) - 24/(18+x) = 1	1.0 M
2	Forming quadratic equation x^2 + 48x - 324 = 0	1.0 M
3	Solving to get x = 6 km/h	1.0 M

Q8. (Quadratic Equations) **[3 Marks]**

Solve for

$$x$$

by quadratic formula:

$$4sqrt(3)x^2 + 5x - 2sqrt(3) = 0$$

.

Model Answer:

Here

$$a = 4sqrt(3), b = 5, c = -2sqrt(3)$$

. Discriminant

$$D = b^2 - 4ac = 25 - 4(4sqrt(3))(-2sqrt(3)) = 25 + 96 = 121$$

. Then

$$x = \frac{-b + -\sqrt{D}}{2a} = \frac{-5 + -11}{8sqrt(3)}$$

. Root 1:

$$x = \frac{6}{8sqrt(3)} = \frac{3}{4sqrt(3)} = \frac{\sqrt{3}}{4}$$

. Root 2:

$$x = \frac{-16}{8sqrt(3)} = -\frac{2}{\sqrt{3}} = -\frac{2sqrt(3)}{3}$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Calculating D = 121	1.0 M
2	Applying quadratic formula	1.0 M
3	Finding roots sqrt(3)/4 and −2/sqrt(3)	1.0 M

SECTION C (Short Answer Questions - 3 Marks each)

Q9. (Quadratic Equations) **[5 Marks]**

Two water taps together can fill a tank in

$$9\frac{3}{8} \text{ hours} = \frac{75}{8} \text{ hours}$$

. The tap of larger diameter takes

$$10 \text{ hours}$$

less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

Model Answer:

Let smaller tap take

$$x$$

hours. Then larger tap takes

$$(x-10)$$

hours. Combined rate:

$$\frac{1}{x} + \frac{1}{x-10} = \frac{8}{75} \implies \frac{2x-10}{x(x-10)} = \frac{8}{75} \implies 75(2x-10) = 8(x^2-10x) \implies 150x - 750 = 8x^2 - 80x \implies 8x^2 - 230x + 750 = 0 \implies 4x^2 - 115x + 375 = 0$$

. Factoring:

$$(x-25)(4x-15) = 0 \implies x = 25$$

or

$$x = 3.75$$

. If

$$x = 3.75$$

,

$$x - 10$$

is negative, which is impossible. Hence smaller tap takes

$$25 \text{ hours}$$

and larger tap takes

$$15 \text{ hours}$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Setting up fractional work equation 1/x + 1/(x-10) = 8/75	1.5 M
2	Simplifying to standard quadratic form 4x^2 - 115x + 375 = 0	1.5 M
3	Factoring and rejecting inadmissible root x = 3.75	1.0 M
4	Stating final answer: 25 hours and 15 hours	1.0 M

Q10. (Quadratic Equations) **[4 Marks]**

A rectangular park of length

$$L$$

and breadth

$$B$$

has an area of

$$528\ \text{m}^2$$

. The length of the plot is one more than twice its breadth. (i) Formulate the quadratic equation representing this situation. [1 Mark] (ii) Find the length and breadth of the park by factoring. [2 Marks] (iii) Calculate the cost of fencing the park at

$$\text{Rs. } 25 \text{ per meter}$$

. [1 Mark]

Model Answer:

(i) Let breadth

$$B = x\ \text{m}$$

. Length

$$L = (2x + 1)\ \text{m}$$

. Area

$$x(2x + 1) = 528 \Longrightarrow 2x^2 + x - 528 = 0$$

. (ii) Factoring:

$$2x^2 + 33x - 32x - 528 = x(2x + 33) - 16(2x + 33) = (2x + 33)(x - 16) = 0 \Longrightarrow x = 16\ \text{m}$$

. Breadth

$$= 16\ \text{m}$$

, Length

$$= 2(16) + 1 = 33\ \text{m}$$

. (iii) Perimeter

$$= 2(L + B) = 2(33 + 16) = 98\ \text{m}$$

. Cost

$$= 98 \times 25 = \text{Rs. } 2450$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Setting up 2x^2 + x - 528 = 0	1.0 M
2	Factoring to get x = 16 m and length = 33 m	2.0 M
3	Calculating perimeter 98 m and cost Rs. 2450	1.0 M

SECTION D (Long Answer Questions - 5 Marks each)

Q11. (Quadratic Equations) **[5 Marks]**

An express train takes

$$1 \text{ hour}$$

less than a passenger train to travel

$$132 \text{ km}$$

between Mysore and Bangalore. If the average speed of the express train is

$$11 \text{ km/h}$$

more than that of the passenger train, find the average speed of the two trains.

Model Answer:

Let passenger train speed be

$$x \text{ km/h}$$

. Express train speed is

$$(x + 11) \text{ km/h}$$

. Time difference:

$$\frac{132}{x} - \frac{132}{x+11} = 1 \implies 132\left(\frac{11}{x(x+11)}\right) = 1 \implies x^2 + 11x - 1452 = 0$$

. Factoring:

$$(x + 44)(x - 33) = 0 \implies x = 33$$

(speed cannot be negative). Average speed of passenger train

$$= 33 \text{ km/h}$$

and express train

$$= 44 \text{ km/h}$$

.

Step-Wise Marking Breakdown:

Step	Evaluation Criteria / Step Rule	Marks
1	Setting up time equation 132/x - 132/(x+11) = 1	1.5 M
2	Deriving quadratic equation x^2 + 11x - 1452 = 0	1.5 M
3	Factoring and finding x = 33	1.0 M
4	Stating speeds: 33 km/h and 44 km/h	1.0 M